NERVE AGENT EXPOSURE: ALS AND BLS
MARK I AUTOINJECTOR USE BY EMERGENCY MEDICAL TECHNICIAN AND PARAMEDICS
O-1, Reviewed June 1, 2017

The Mark I Autoinjector contains antidotes to be used in instances of exposure to a nerve agent. It consists of two autoinjectors containing Atropine Sulfate 2 mg and Pralidoxime Chloride. (2-PAM CL) 600 mg.

Atropine is the primary drug for the treatment of nerve agent exposure. It relieves smooth muscle constriction in the lungs (wheezing, respiratory distress) and GI (diarrhea, cramps) tract, and also dries up respiratory tract secretions. Atropine administered by the Mark I Auto-Injector will show some effectiveness in 3-5 minutes.

2-PAM CL is effective at re-establishing normal skeletal muscle contraction (relieves twitching and paralysis of the respiratory muscles).

The CANA Injector contains 10 mg of Valium to be used for seizure control. **ALS ONLY**

Auto-Injector Use:
The Mark I Autoinjector may be administered by qualified emergency medical technicians and paramedics who have had training in site recognition and treatment of nerve agent intoxication in the event of a chemical release. **The Emergency Medical Technician shall only administer the Mark I Auto-Injector under the direct authority of a paramedic.**

Use of the Mark I injector is to be based on signs and symptoms of the patient.
These will include:

SLUDGEM + Muscle twitching or spasm + Bradycardia, SOA, Respiratory distress + confusion, agitation, seizures, or coma

The suspicion or the identified presence of a nerve agent is not sufficient reason to administer these medications. Medical treatment of these exposures is directed to relieving respiratory distress and alleviating seizure.

NOTE: These injectors are not to be used as a prophylactic modality. There is to be no self-administration of the antidote.

CAUTION IS TO BE EXERCISED AT ALL TIMES. IN THE INITIAL PHASE, TRIAGE WILL BE INITIATED IN THE HOT ZONE AND CONTINUED IN THE WARM ZONE BY PERSONNEL IN APPROPRIATE CPE. PATIENT DECONTAMINATION WILL BE SIMULTANEOUS WITH AND/OR PRIOR TO TREATMENT. CHILDREN SHOULD BE DECONTAMINATED AND HAVE EXPEDITED TRANSPORT OFF THE SCENE ESPECIALLY IF THEY ARE DEMONSTRATING ANY SIGNS OR SYMPTOMS OF EXPOSURE.

General Guidelines for Treatment:

1. In all cases, remove secretions, maintain patient's airway, and if necessary, and the situation permits, apply oxygen and bag mask.

2. Atropine must be given first. See dosing charts.

3. If symptoms resolve, then only monitoring is necessary.

4. 2-PAM CL is most effective if administered immediately after the poisoning, but not before Atropine, especially for severe exposures.

5. ***ALS ONLY***: Diazepam may be cautiously given if convulsions are not controlled.

6. ***ALS ONLY***: Xopenex 1.25 mg via nebulization may be used for any patient who presents with bronchospasm secondary to nerve agent exposure after antidote administration.

7. ***ALS ONLY***: Atrovent unit dose via nebulization may also be administered to assist in drying secretions in any patient who presents with bronchospasm secondary to nerve agent exposure after antidote administration.

8. When the nerve agent has been ingested, exposure may continue for some time due to slow absorption from the lower bowel, and fatal relapses have been reported after initial improvement. Continued medical monitoring and transport is mandatory.

9. If dermal exposure has occurred, decontamination is critical and should be done with standard decontamination procedures. Patient monitoring should be directed to the same signs and symptoms as with all nerve agents.

10. Eyes must be decontaminated within minutes of exposure to limit injury. Flush for 5-10 minutes. There is no need to flush the eyes following exposure to nerve agent vapor. Do not cover the eyes with bandages.

11. Documentation shall be done on triage tags. Each dose of Atropine should be recorded with an "A", each dose of 2-PAM CL with a "P", and each dose of Valium with a "D" or "V".

12. If patients continue to exhibit signs and symptoms of nerve agent intoxication, they may continue to receive Atropine and 2-PAM CL until symptoms cease or maximum dosing has been achieved.

NOTE: MAXIMUM DOSE OF 2-PAM CL IS 1.8 GRAMS

ATROPINE AND 2-PAM CL AUTO-INJECTOR IS NOT TO BE USED IN CHILDREN UNDER 10 YEARS OF AGE SEE PEDIATRIC DOSING CHART FOR ADMINISTRATION BY PARAMEDICS ONLY.

THERE IS NO MAXIMUM DOSE OF ATROPINE

Initial Antidote Dosing Schedule BLS

Signs and symptoms	Atropine Dose Monitor Interval	2-PAM CL
SLUDGEM + Severe respiratory distress + Agitation and Seizures	3 Auto-Injectors (6 mg) in rapid succession Monitor every five minutes	3 Auto-Injectors (1.8 Gm) in rapid succession
SLUDGEM + Respiratory distress	2 Auto Injectors (4 mg) Monitor every five minutes	2 Auto Injectors (1200mg)
Pinpoint pupils and sudden onset of nasal discharge, No shortness of breath	1 Auto-Injector (2 mg) Monitor every five minutes	None
Asymptomatic or pinpoint pupils ONLY	None Monitor for signs and symptoms every fifteen minutes	None

Initial Antidote Dosing Schedule: ALS
May follow BLS dosing

Signs and Symptoms	Atropine Dose Monitoring Interval	2 PAM CL	Diazepam CANA Injector
SLUDGEM + Severe Respiratory Distress + Agitation + Seizures	Atropine 2 mg IV/IM every 5-10 minutes until secretions dry, and SOA improves See Pediatric Dosing age < 10	3 Auto-Injectors or 1,800 mg IM Elderly: 25 mg/Kg IM See Pediatric Dosing age < 10	1 CANA Injector (10 mg) IM) OR 5-10 mg IV at 2 mg/minute after administration of atropine and 2 PAM CL. Repeat as needed for seizure control May follow seizure protocol:
SLUDGEM + Respiratory distress	Atropine 2 mg IV/IM every 5-10 minutes until secretions dry, and SOA improves Elderly: May begin at 1 mg IM See Pediatric Dosing age < 10	2 Auto-Injectors or 1200 mg IM Elderly: 10 mg/KG IM See Pediatric Dosing age < 10	None unless seizing. If seizing follow severe dosing regime
Pinpoint pupils and sudden nasal discharge No SOA	Atropine 1 mg x 1 IV/IM See Pediatric Dosing age < 10	None	None
Asymptomatic or pinpoint pupils only	None Monitor for signs and symptoms every fifteen minutes	None	None

10. Once patients have been initially managed and patient volume allows for more protracted assessments you may:

Follow-up dosing schedule:

Signs and Symptoms	Atropine Repeat Dosing Frequency
SLUDGEM + Agitation + Severe Respiratory Distress	Atropine one Auto-Injector (2 mg) every 3-5 minutes until SOA relieved/secretions drying
SLUDGEM + Respiratory Distress	Atropine one Auto-injector (2 mg) every 5-10 minutes until SOA is relieved
Pinpoint pupils and sudden onset of nasal discharge No SOA	Monitor for worsening signs and symptoms every ten minutes. Repeat atropine as needed.
Asymptomatic or pinpoint pupils ONLY	Observe and treat as outlined

In the event EMS Personnel become exposed to a Nerve Agent and they meet the above criteria, decontaminate and the Mark I Auto-Injector may be administered.

Directions for use of the Mark I Auto-Injector:

1. First remove the smaller of the two injectors (Atropine) out of the clip and remove the safety cap. Do not touch the colored end of the injector after removing the cap. CAUTION: The injector can and will inject into a finger or hand if any pressure ia applied to either end on the injector. This will not deliver an accurate dose of medication.

2. Hold injector as you would a pen. Place colored end on thickest part of the thigh (Vastus lateralis muscle) and press hard until injector is activated.

3. Hold firmly in place for ten seconds and then remove. The time allows the fluid to be injected.

4. If called for, remove the larger injector from the kit (2-PAM CL) and remove the safety cap. Follow the same procedure for administration as above.

5. Massage the muscle if time allows.

6. Dispose of auto-injector properly in bio-hazard container.

7. Begin decontamination procedures immediately after the first Mark I Auto- Injector kit if it has not already been initiated.

General Cautions:

1. Delayed symptoms may present anytime post incident.

2. Any patient ill enough to receive even one dose of atropine must be evaluated at an appropriate facility.

3. Signs and symptoms of nerve agent poisoning may reappear. Serial observations are a critical part of the management process.

Adverse Reactions:

Atropine reactions may occur BUT THERE ARE NO CONTRAINDICATIONS TO TREATING SYMPTOMATIC PATIENTS.

* Atropine may cause chest pain. It may also exacerbate angina or induce a myocardial infarction.
* Up to 1 hour after IM injection of 2-PAM CL some pain may be experienced at the site of injection.
* 2 PAM CL may cause blurred vision, double vision, dizziness, headache, drowsiness, nausea, tachycardia, increased blood pressure, and hyperventilation.
* Both (Atropine and 12-PAM CL) should be used with caution (but not withheld) in patients with preexisting cardiac disease, high blood pressure, or strokes, particularly in the management phase.

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Nerve Agents in Children: Guidelines

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Symptoms	Triage Level: Disposition	Atropine Correct hypoxia before IV use (risk of torsades, Vfib)	Pralidoxime	Diazepam May use other benzodiazepines (e.g. midazolam)
Asymptomatic	**Delayed:** Observe	None	None	None
Miosis, mild rhinnorhea	**Delayed:** Admit or Observe prn	None	None	None
Miosis and any other symptom	**Immediate** -Moderate: Admit	**0.05 mg/kg IV or IM** • repeat as needed q5-10 minutes until respiratory status improves	**25-50 mg/kg IV or IM**, may repeat q 1 hour. • **Watch for:** ⇒ muscle rigidity ⇒ laryngospasm, ⇒ tachycardia	**For any neurologic effect:** • **30 days to 5 years – 0.05 to 0.3 mg/kg IV** to a max of 5mg/dose. • **5 years and older– 0.05 to 0.3 mg/kg IV** to a max of 10 mg/dose. May repeat q15-30 minutes
Apnea, Convulsions, Cardiopulmonary Arrest	**Immediate** - Severe: Admit intensive care status	**0.05-0.1 mg/kg IV, IM, per ETT** • **no maximum** • repeat q5-10 minutes as above	**25-50 mg/kg IV or IM as above**	**See above**

Consider other supportive agents as indicated: Oxygen, Bronchodilalors, Analgesics, Mydriatics, Environmental protection